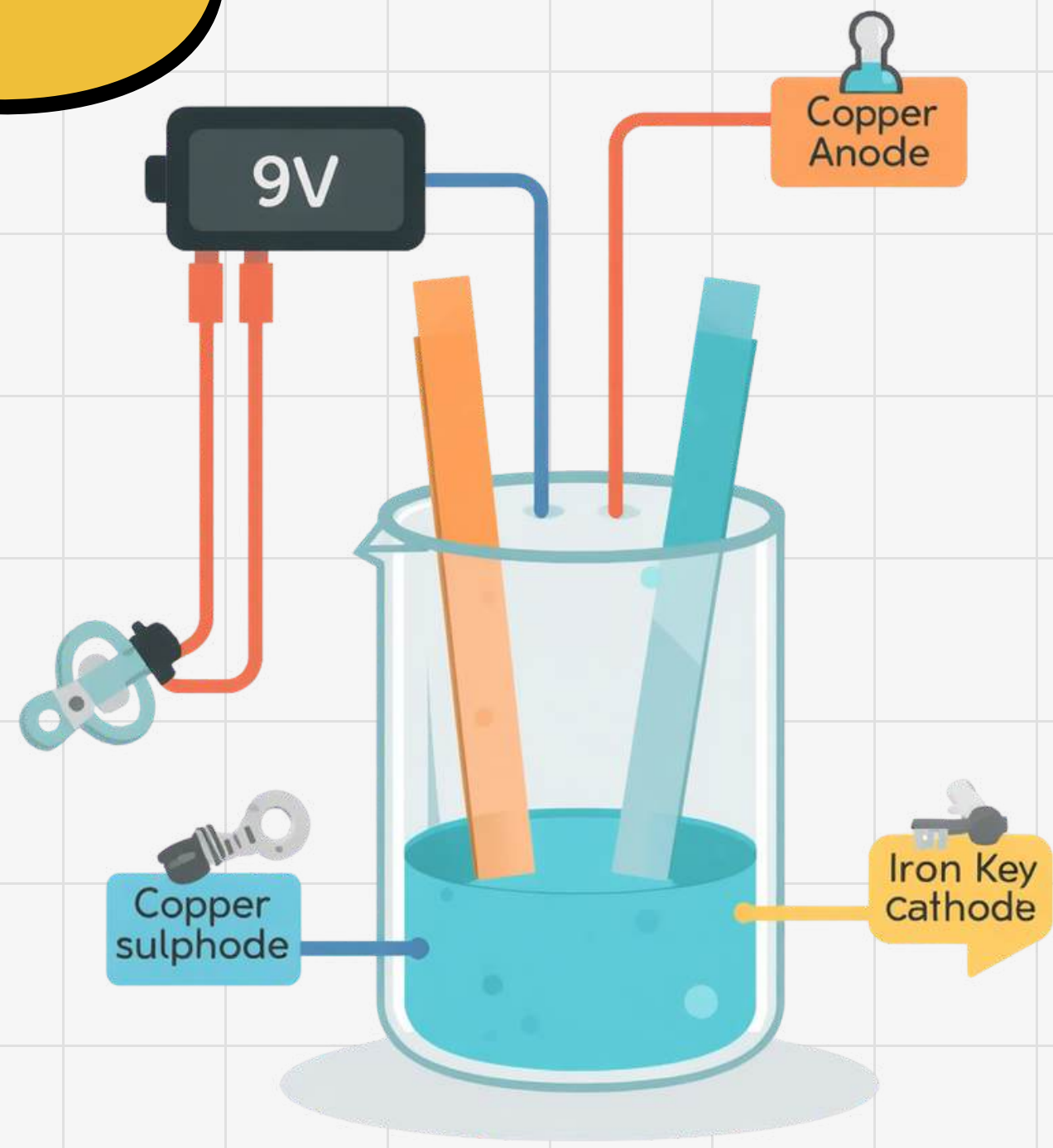


Electroplating Magic!

Why does one key look dull and the other shiny?





What Is Electroplating?

Electroplating means putting a thin shiny layer of metal on another object using electricity!

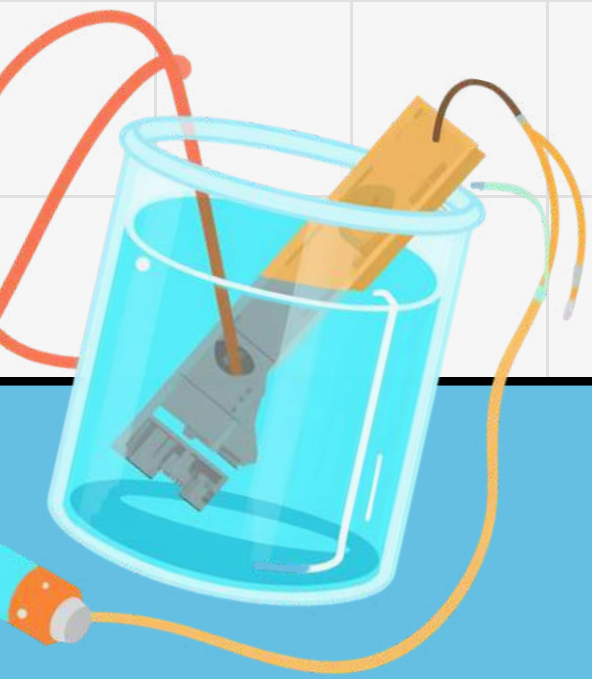
- ⚡ Anode = Copper strip (copper giver)
- 🔑 Cathode = Iron key (key receiver)
- 💧 Solution = Neela Thotha (blue copper sulphate)

Safety First: Jugaad Safety Tips!

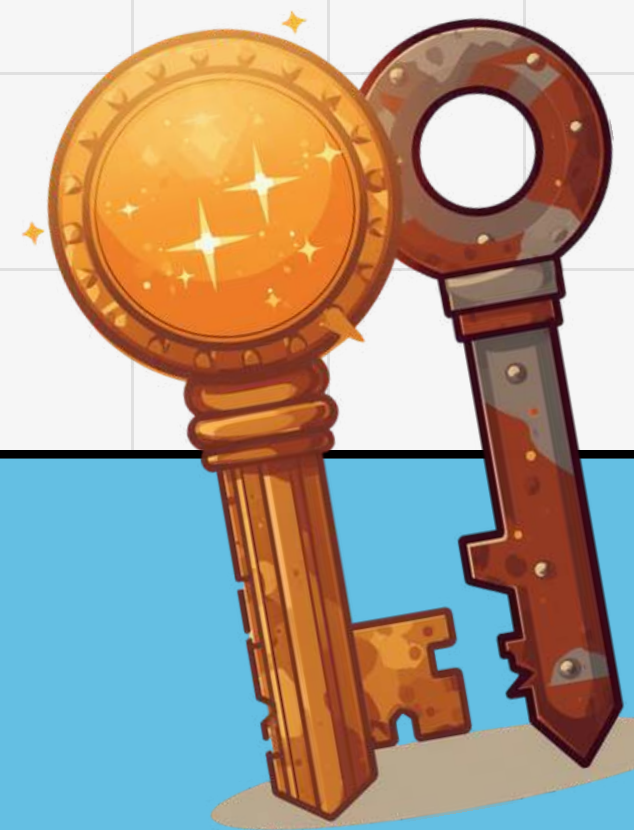
- 🧤 Wear gloves & goggles to protect hands and eyes
- 🔋 Use only a low-voltage 9V battery (no wall plugs!)
- ⚠️ Handle Copper Sulphate carefully — no tasting or touching!
- 🧹 Keep your workspace clean and dry at all times



How to Copper Coat a Key: Step-by-Step



1. 🛠️ Clean the key to remove rust and dirt
2. 💧 Prepare the blue Neela Thotha solution in a small container
3. + Connect copper strip (anode) to positive (+) battery wire
4. - Connect iron key (cathode) to negative (-) battery wire
5. 🧴 Dip both into the solution without touching each other
6. ✨ Wait and watch a shiny copper coat form on the key!



Ue



Electroplating in Real Life!



Gold-plated temple ornaments & Puja Kalash



Chrome-plated bicycle & door handles



Shiny brass utensils & taps



Faux jewellery with copper coating

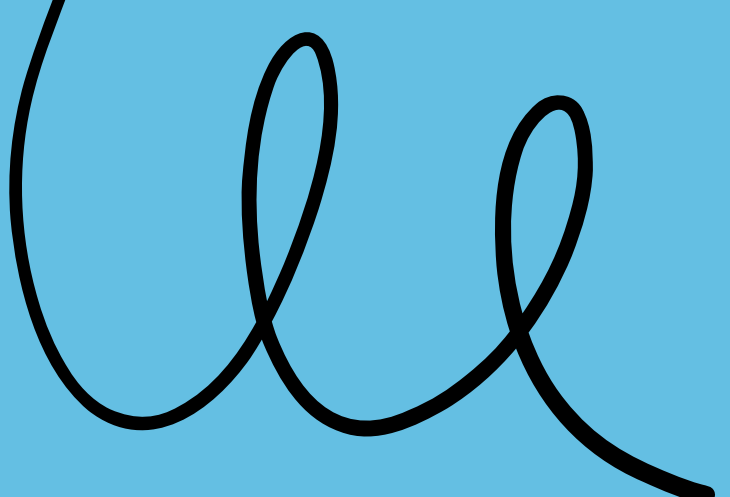
Pop Quiz!



Which part of the setup gives copper to the key?

- A) The iron key
- B) The copper strip ✓
- C) The 9V battery

Raise your hand if you know!



Let's Remember!

- ⚡ Electroplating: putting a shiny layer of metal using electricity
- ◆ Anode (Copper Giver): copper strip that loses copper
- 🔑 Cathode (Key Receiver): iron key that gets copper coating
- 💧 Copper Sulphate: blue copper sulphate solution used for plating



Electricity



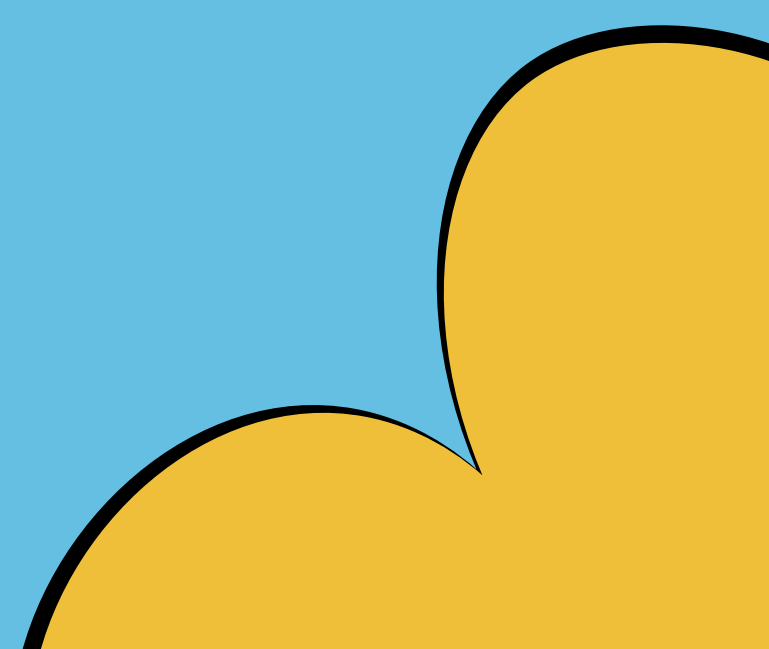
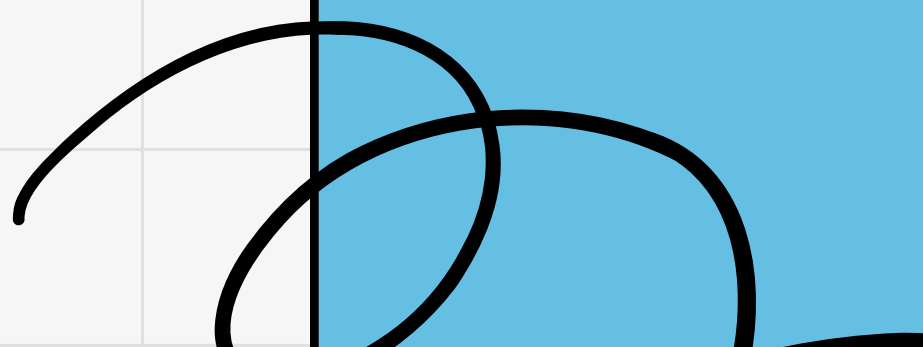
Copper Strip



Iron Key



Blue Jar

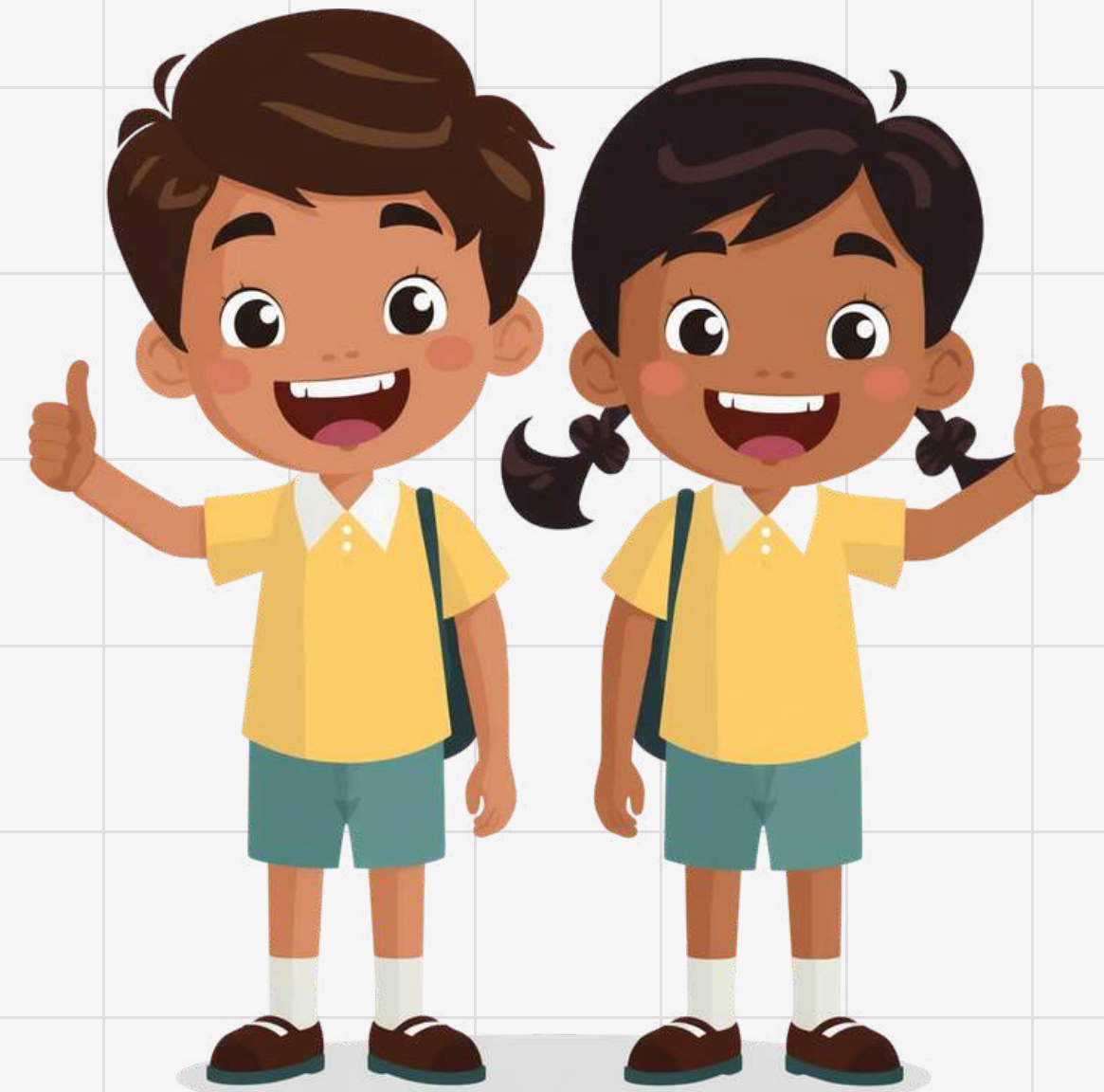


Your Turn!

Why do you think electroplating is better than just painting the key with copper color?

Share your answer aloud or write it down!

Great job! Keep exploring science around you! ✨



Thank You !

Any Questions ! See you
in the next session

